Dataset Overview

The data utilized in this report is a simulated, multi-domain clinical trial dataset consisting of 500 patients, alongside associated pharmacokinetic and facility operational data. The data is structured into three distinct data frames to support a comprehensive suite of statistical modeling techniques.

1. Clinical Trial Data (clinical_data)

This is the primary dataset used for cross-sectional and survival analysis, containing individual patient demographics, baseline metrics, and treatment outcomes.

patient_id: A unique identifier for each of the 500 patients.
age: Patient age in years, normally distributed around a mean of 60.
treatment_group: A categorical factor representing the assigned cohort (Placebo, Drug_A, or Drug_B).
baseline_bp: The patient's initial blood pressure reading prior to treatment.
post_bp: The patient's final blood pressure reading after the treatment period.
side_effects: A count variable representing the number of mild side effects reported by the patient during the trial.
complication: A binary indicator variable (1 = Yes, 0 = No) tracking whether the patient experienced a major medical complication.
months_to_relapse: A continuous time-to-event variable measuring the months elapsed until disease relapse or the end of the observation period.
relapse_event: A binary censoring status flag indicating whether the relapse actually occurred (1) or if the patient was censored/did not relapse (0).

2. Pharmacokinetic Data (nl_data)

This dataset tracks the concentration of the experimental drug in the bloodstream over a 24-hour period, used for non-linear modeling.

hour: Time elapsed in hours since the drug was administered (measured at 30-minute intervals).
concentration: The measured concentration of the drug remaining in the blood plasma.

3. Operational Time Series Data (ts_data)

This dataset tracks the administrative load of the clinical facility over a three-year period, used for ARIMA forecasting.

date: Daily chronological index from January 1, 2020, to December 31, 2022.
admissions: The total count of daily hospital admissions, reflecting underlying trends and seasonal operational cycles.

This file includes the statistical interpretation of the outputs of our model

1. ANOVA Model (Analysis of Variance)

```
> summary(anova_model)
                 Df Sum Sq Mean Sq F value Pr(>F)
treatment_group   2  32862   16431   63.43 <2e-16 ***
Residuals       497 128750     259
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

The ANOVA results indicate a highly significant difference in blood pressure between the treatment groups ($F(2, 497) = 63.43$, $p < 2e-16$).

To determine exactly where these differences lie, we applied Tukey's Honest Significant Difference (HSD) test

```
> TukeyHSD(anova_model)
  Tukey multiple comparisons of means
    95% family-wise confidence level

Fit: aov(formula = post_bp ~ treatment_group, data = clinical_data)

$treatment_group
                   diff       lwr        upr p adj
Drug_A-Placebo  -9.99350 -14.09645  -5.890549 1e-07
Drug_B-Placebo -20.23685 -24.46061 -16.013086 0e+00
Drug_B-Drug_A  -10.24335 -14.36657  -6.120130 0e+00
```

Post-Hoc Findings:

- **Drug A vs. Placebo:** Drug A significantly lowers blood pressure by an average of 9.99 points compared to the placebo ($p < 0.001$).
- **Drug B vs. Placebo:** Drug B significantly lowers blood pressure by an average of 20.24 points compared to the placebo ($p < 0.001$).
- **Drug B vs. Drug A:** Drug B is significantly more effective than Drug A, lowering blood pressure by an additional 10.24 points ($p < 0.001$).

2. ANCOVA Model (Analysis of Covariance)

```
> summary(ancova_model)
                 Df Sum Sq Mean Sq F value Pr(>F)
treatment_group   2  32862   16431   626.4 <2e-16 ***
baseline_bp       1 115740  115740  4412.5 <2e-16 ***
Residuals       496  13010      26
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Interpretation:

Including baseline_bp as a covariate drastically improved the model. By absorbing the variance attributed to starting blood pressure, the residual Mean Square error plummeted from 259 (in the ANOVA) to just 26. Consequently, the F-value for our treatment_group spiked to 626.4. Both the treatment group and baseline blood pressure are highly significant predictors ($p < 2e-16$).

3. Generalized Linear Model: Logistic Regression

We utilized a logistic regression model with a logit link function to predict the binary likelihood of a patient experiencing a major complication.

```
Coefficients:
              Estimate Std. Error z value Pr(>|z|)
(Intercept) -4.686e+00  1.423e+00  -3.292 0.000995 ***
age          4.964e-02  1.313e-02   3.782 0.000156 ***
baseline_bp -8.416e-05  8.047e-03  -0.010 0.991655
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 439.67  on 499  degrees of freedom
Residual deviance: 424.68  on 497  degrees of freedom
AIC: 430.68

Number of Fisher Scoring iterations: 4
```

Interpretation:

Age: Age is a highly significant risk factor ($p = 0.000156$). The coefficient is 0.04964. Converting these log-odds to an Odds Ratio (e^0.04964 =1.051), we determine that for every additional year of age, a patient's odds of experiencing a complication increase by roughly 5.1%.
Baseline BP: Interestingly, baseline blood pressure is not a significant predictor of complications in this model ($p = 0.991655$).

4. Generalized Linear Model: Poisson Regression

To analyze the count of mild side effects, we utilized a Poisson regression model with a log link function.

```
Coefficients:
                     Estimate Std. Error z value Pr(>|z|)
(Intercept)          -0.09716    0.08248  -1.178    0.239
treatment_groupDrug_A  1.18830    0.09316  12.755   <2e-16 ***
treatment_groupDrug_B  1.28975    0.09333  13.819   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for poisson family taken to be 1)

    Null deviance: 797.77  on 499  degrees of freedom
Residual deviance: 529.49  on 497  degrees of freedom
AIC: 1714.3

Number of Fisher Scoring iterations: 5
```

5. Non-Linear Modeling (NLS)

We modeled the pharmacokinetics of the drug clearing the bloodstream using a non-linear least squares exponential decay function: C=C0.e^-kt .

```
Formula: concentration ~ C0 * exp(-k * hour)

Parameters:
    Estimate Std. Error t value Pr(>|t|)
C0 1.001e+02  1.230e+00   81.42   <2e-16 ***
k  1.536e-01  2.593e-03   59.24   <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.035 on 46 degrees of freedom

Number of iterations to convergence: 5
Achieved convergence tolerance: 1.819e-07
```

Interpretation:

The optimization algorithm successfully converged in 5 iterations, establishing highly precise parameter estimates ($p < 2e-16$ for both):

- **Initial Concentration (C_0):** Estimated at 100.1 units.
- **Elimination Rate (k):** Estimated at 0.1536, meaning the drug clears the system at a rate of approximately 15.36% per hour.
- **Model Fit:** The residual standard error is 2.035, indicating that the model's predicted concentration curve fits the actual data points very closely.

6. Time series Modeling (ARIMA)

To forecast future daily hospital admissions, we trained an Auto-Regressive Integrated Moving Average (ARIMA) model on our time series data.

```
Series: admissions_ts
ARIMA(5,1,2)

Coefficients:
          ar1      ar2      ar3      ar4      ar5      ma1      ma2
      -0.6711  -0.0967  -0.0770  -0.1139  -0.0910  -0.2793  -0.4494
s.e.   0.3073   0.0603   0.0499   0.0473   0.0349   0.3074   0.2635

sigma^2 = 27.39:  log likelihood = -3363.31
AIC=6742.63   AICc=6742.76   BIC=6782.62

Training set error measures:
                      ME     RMSE      MAE        MPE     MAPE      MASE
Training set -0.06496994 5.214294 4.151906 -0.9006446 7.147131 0.1575664
                      ACF1
Training set -0.0009068381
```

Interpretation:

The auto.arima algorithm determined that an **ARIMA(5,1,2)** model provided the optimal fit for the data.

- **Model Structure:** The data required one order of differencing (d=1) to achieve stationarity. The model relies heavily on past values, using 5 autoregressive terms (p=5) and 2 moving average error terms (q=2) to make predictions.
- **Error Metrics:** On the training set, the Root Mean Squared Error (RMSE) is 5.214. This indicates that the model's historical daily admission predictions are, on average, off by roughly 5.2 patients.

7. Survival Analysis(Cox Proportional-Hazards)

```
  n= 500, number of events= 367

                           coef exp(coef)  se(coef)      z Pr(>|z|)
treatment_groupDrug_A -0.929343  0.394813  0.132783 -6.999 2.58e-12 ***
treatment_groupDrug_B -0.806998  0.446196  0.135728 -5.946 2.75e-09 ***
age                    0.003134  1.003138  0.005692  0.551    0.582
baseline_bp           -0.004727  0.995284  0.003420 -1.382    0.167
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

                      exp(coef) exp(-coef) lower .95 upper .95
treatment_groupDrug_A    0.3948     2.5328    0.3043    0.5122
treatment_groupDrug_B    0.4462     2.2412    0.3420    0.5822
age                      1.0031     0.9969    0.9920    1.0144
baseline_bp              0.9953     1.0047    0.9886    1.0020

Concordance= 0.61  (se = 0.017 )
Likelihood ratio test= 51.51  on 4 df,   p=2e-10
Wald test            = 56.42  on 4 df,   p=2e-11
Score (logrank) test = 59.49  on 4 df,   p=4e-12
```

Interpretation:

The model analyzed 500 patients, observing 367 relapse events.

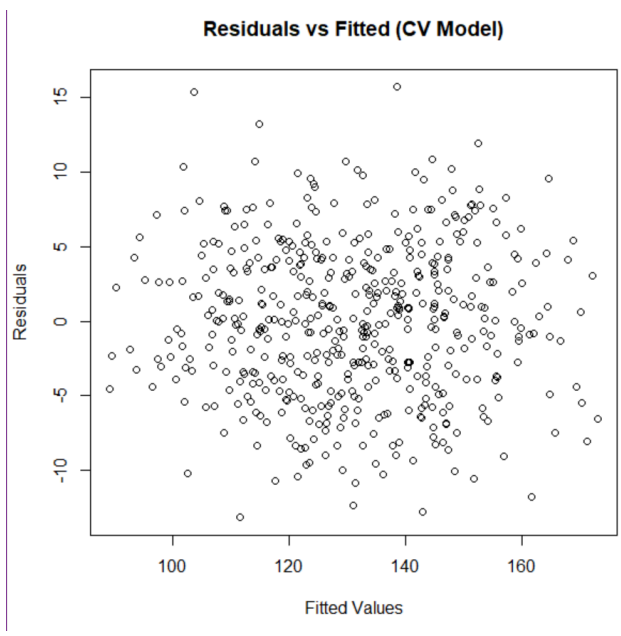
- **Treatment Effectiveness:** Both experimental drugs significantly delay relapse compared to the placebo.
    - **Drug A:** The Hazard Ratio (exp(coef)) is 0.3948 (p = 2.58e-12). This means patients on Drug A have roughly 39.5% of the relapse hazard of the placebo group, effectively reducing the risk of relapse by 60.5%.

    - **Drug B:** The Hazard Ratio is 0.4462 (p = 2.75e-09). Patients on Drug B have roughly 44.6% of the relapse hazard compared to the placebo.
- **Other Covariates:** Interestingly, neither patient age (p = 0.582) nor baseline blood pressure (p = 0.167) proved to be statistically significant risk factors for the time to relapse in this model.
- **Overall Accuracy:** The model achieved a Concordance index of 0.61, indicating a moderate ability to correctly rank patient survival times.

8. Model Validation and Diagnostics

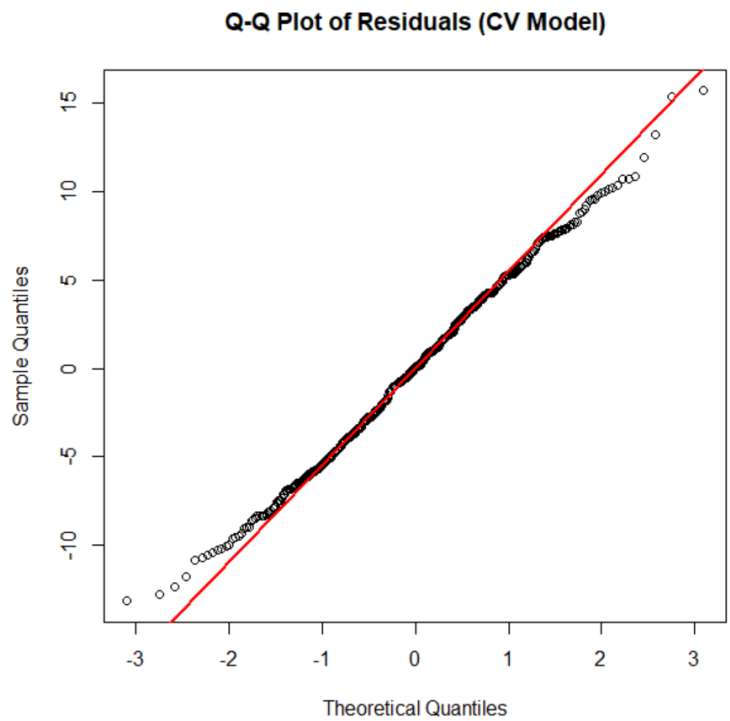
To ensure the mathematical validity of our linear models (specifically the Cross-Validated ANCOVA), we evaluated the residual diagnostic plots.

Linearity and Homoscedasticity:
The Residuals vs Fitted plot confirms that our model meets the core assumptions of linearity and equal variance. The model's residuals are scattered randomly around the zero line, resembling a "night sky" with no discernable U-shapes or funneling (megaphone) patterns.



Independence of Errors: An evaluation of the residuals plotted against the data index reveals no sequential or temporal patterns. The model's errors appear completely random from the first patient observation to the last, satisfying the statistical assumption of independence.

Normality of Residuals: The true Normal Q-Q plot demonstrates that the model's residuals are normally distributed. The plotted sample quantiles tightly hug the red 45-degree theoretical reference line, with only negligible, expected deviations at the extreme tails.



Diagnostic Conclusion: Because all core mathematical assumptions are satisfied, we can confidently rely on the p-values, confidence intervals, and predictive metrics generated by our modeling pipeline.